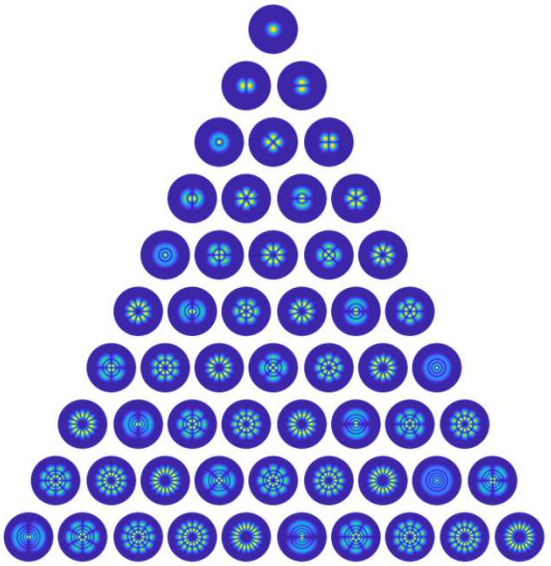
University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications

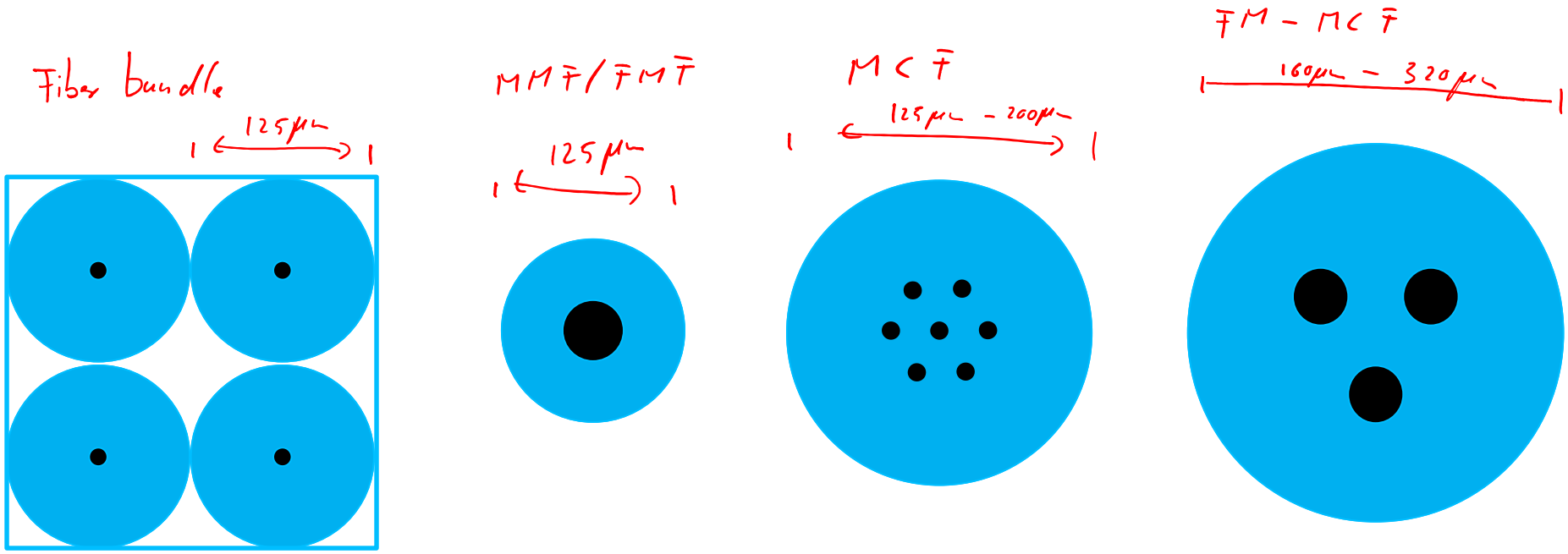
Wintersemester 2025/26

Lecture 9



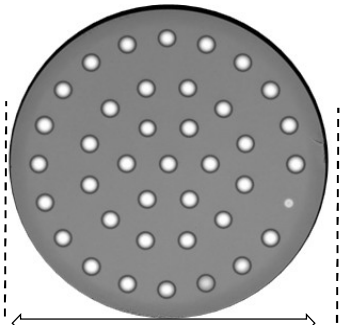
Prof. Dr.-Ing. Georg Rademacher

Optical fibers for space-division multiplexing



Record SDM fiber demonstrations

13 km 38-few-mode multi-core fiber (MCF)

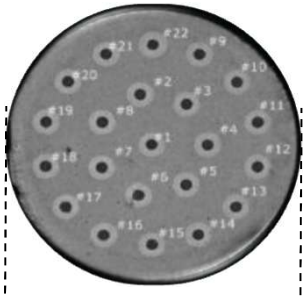


305 µm

2020
10.66 Pb/s
(C+L band)

2023
22 Pb/s
(S/C/L band)

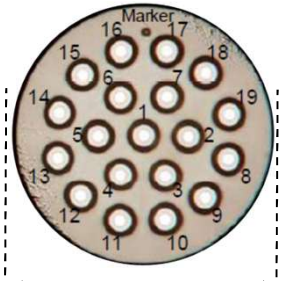
31.4 km 22-core MCF



260 µm

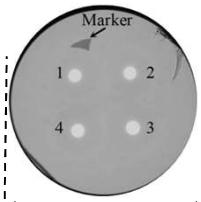
2015
2.15 Pb/s

30 km 19-core MCF



220 µm

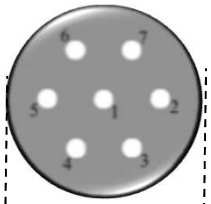
3.5 km 4-core few-mode MCF.



160 µm

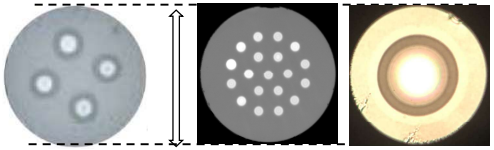
2018
1.2 Pb/s

53.7 km 7-core MCF



160 µm

54-100 km 4-core MCF



125 µm

2022
1.02 Pb/s

66 km randomly coupled MCF

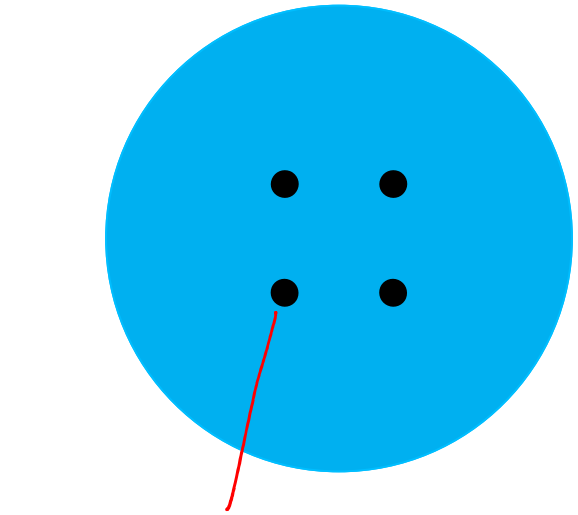
2023
1.7 Pb/s

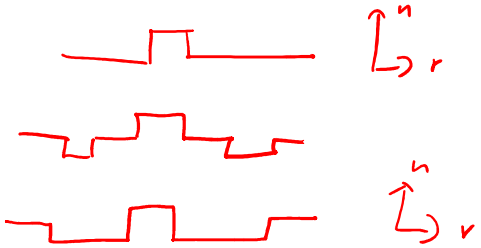
26 km 55-mode fiber

2023
3.56 Pb/s

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany





Design and fabrication of ultra-low crosstalk and low-loss multi-core fiber

Tetsuya Hayashi,* Toshiki Taru, Osamu Shimakawa, Takashi Sasaki, and Eisuke Sasaoka

Optical Communications R&D Laboratories, Sumitomo Electric Industries, Ltd., 1 Taya-cho, Sakae-ku, Yokohama, 244-8588 Japan
**t-hayashi@sei.co.jp*

Multi-core fiber design and analysis: coupled-mode theory and coupled-power theory

Masanori Koshiba,[1,*] Kunimasa Saitoh,[1] Katsuhiro Takenaga[2], and Shoichiro Matsuo[2]

[1]*Division of Media and Network Technologies, Hokkaido University, Sapporo 060-0814, Japan*
[2]*Optics and Electronics Laboratory, Fujikura Ltd., Sakura 285-8550, Japan*
**koshiba@ist.hokudai.ac.jp*

- Biggest trouble: Crosstalk!

Research Article | Vol. 25, No. 10 | 15 May 2017 | OPTICS EXPRESS 12020
Optics EXPRESS

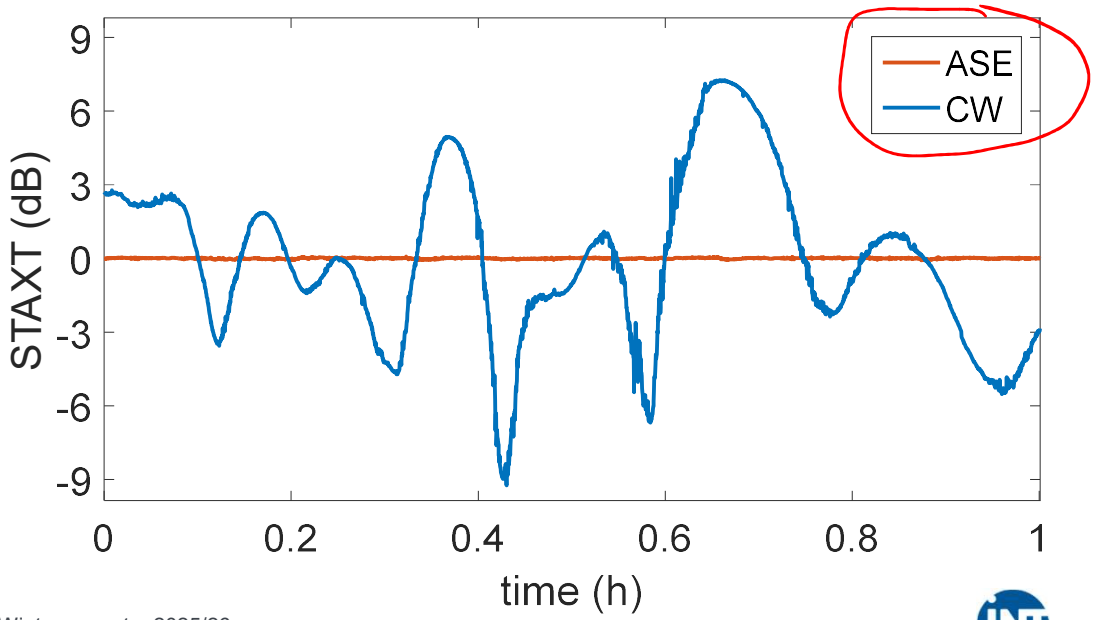
University of Stuttgart
Germany

Crosstalk dynamics in multi-core fibers

GEORG RADEMACHER,* RUBEN S. LUÍS, BENJAMIN J. PUTTNAM, YOSHINARI AWAJI, AND NAOYA WADA

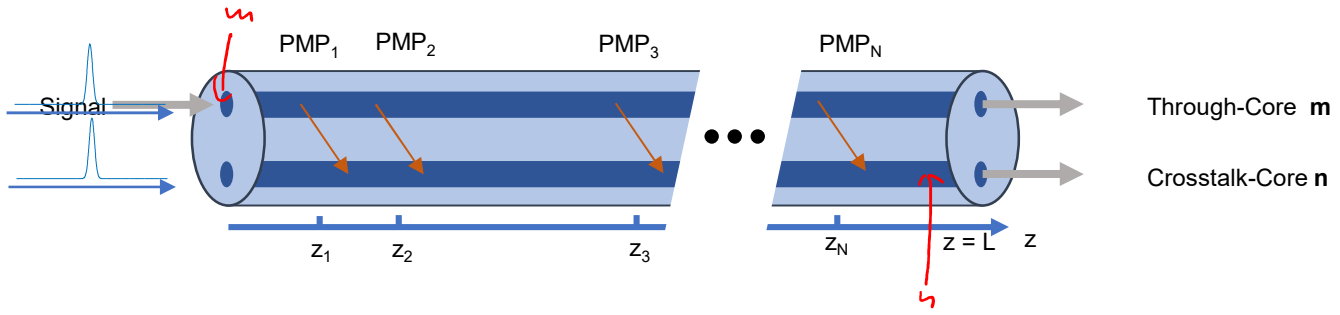
Photonics Systems Research Laboratory, National Institute of Information and Communications Technology (NICT), 4-2-1, Nukui-Kitamachi, Koganei, Tokyo, 184-8795, Japan
**georg.rademacher@nict.go.jp*





Modeling of inter-core crosstalk

University of Stuttgart
Germany



$$A_n(L,\omega) = H_{xT}(L,\omega) A_m(0,\omega)$$

$$H_{xT}(L,\omega) = -j\, k_{nm} \sum_{k=1}^{N} e^{-j\phi_k}\, e^{-j S_{nm} z_k \cdot \omega}$$

T. Hayashi et al., “Behavior of Inter-Core Crosstalk as a Noise and Its Effect on Q -Factor in Multi-Core Fiber,” IEICE Trans. Commun., 2014

R. Luís et al., “Time and Modulation Frequency Dependence of Crosstalk in Homogeneous Multi-Core Fibers,” *JLT,* 2016

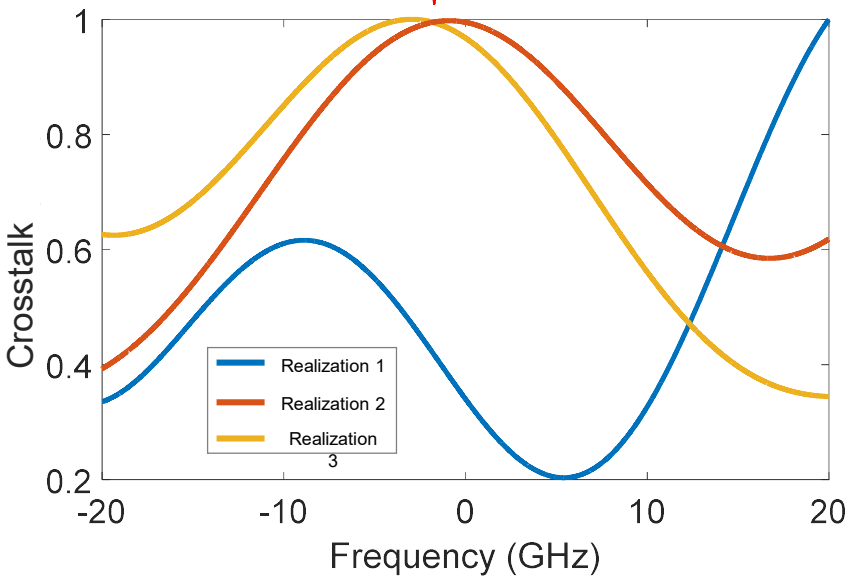
Influence of random phase / inter-core skew on transfer function

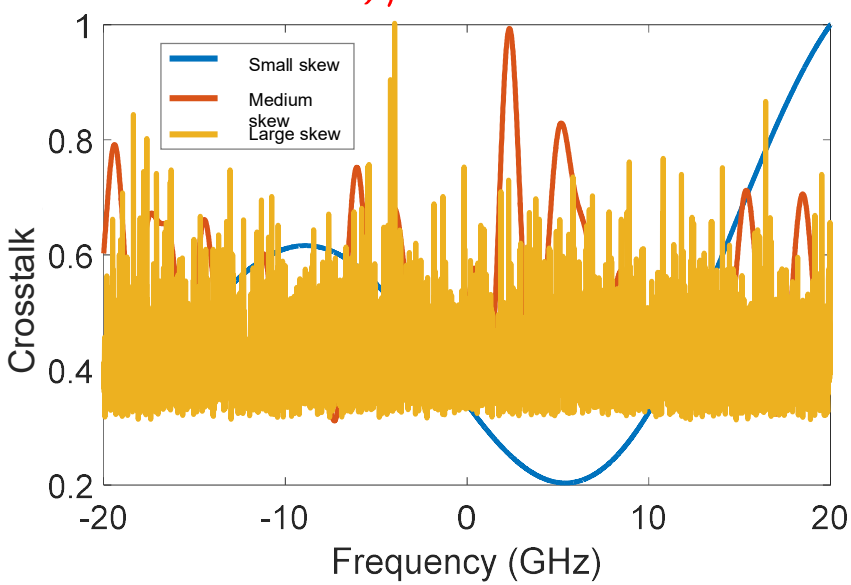
University of Stuttgart
Germany

$$H_{XT}(L,\omega) = -jK_{nm}\sum_{k=1}^{N} e^{-j\phi_k} e^{-j s_{nm} z_k \omega}$$

random phase

different skew





Variance of average crosstalk-power

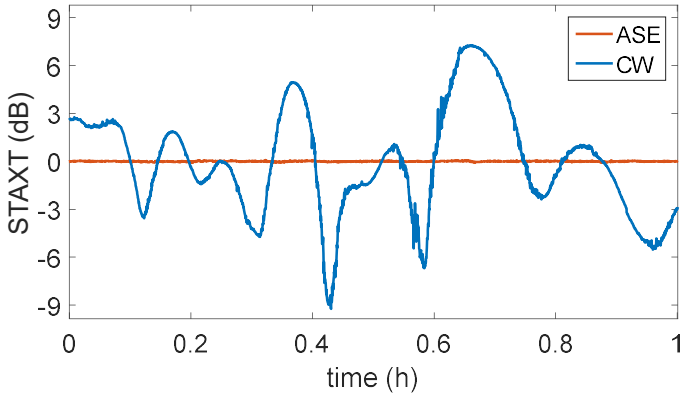
University of Stuttgart
Germany

$$H_{XT}(L,\omega) = -jK_{nm}\sum_{k=1}^{N} e^{-j\phi_k} e^{-js_{nm}z_k\omega}$$

$$VAR_{xT} \propto VAR\left(\int_{-\pi f_{mod}}^{\pi f_{mod}} |H_{xT}(L,\omega)|^2 d\omega\right)$$

$$\propto VAR\left(\sum_{\ell=1}^{M} \underbrace{\left|\sum_{k=1}^{N} e^{-j\phi_k} e^{-j s_{nm} z_k \omega}\right|^2}_{R_\ell}\right)$$

$$VAR_{xT} \propto VAR\left(\sum_{\ell=1}^{M} R_\ell\right) \rightarrow VAR_{xT} \propto \sum_{\ell=1}^{M} VAR(R_\ell) + \sum_{\ell\neq p}^{M}\sum_{p\neq\ell}^{M} cov(R_\ell, R_p)$$

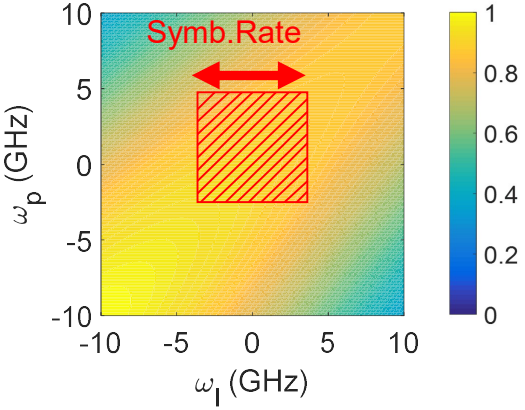


Variance of average crosstalk-power

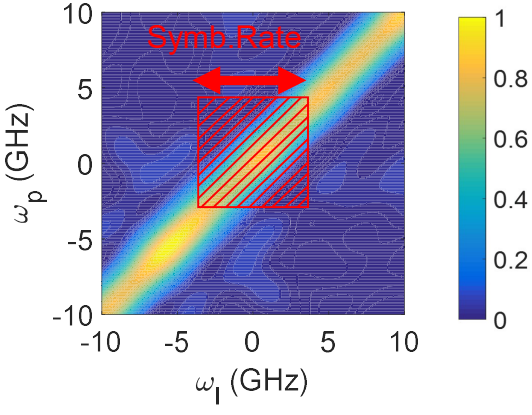
University of Stuttgart
Germany

$$\mathrm{VAR}_{\mathrm{XT}} \propto \sum_{l=1}^{M} \mathrm{VAR}(\boldsymbol{\Omega_l}) + \sum_{l \neq p}^{M} \sum_{p \neq l}^{M} \mathrm{COV}(\boldsymbol{\Omega_l}, \boldsymbol{\Omega_p})$$

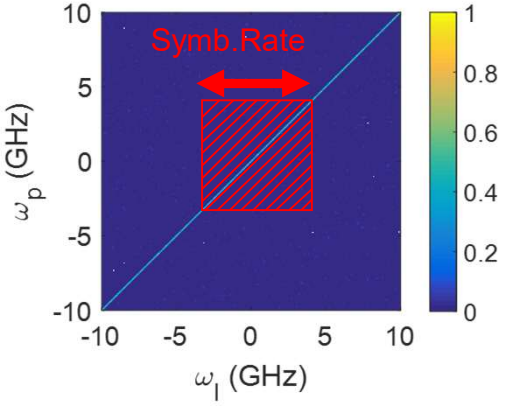
0.025 ns skew



0.25 ns skew



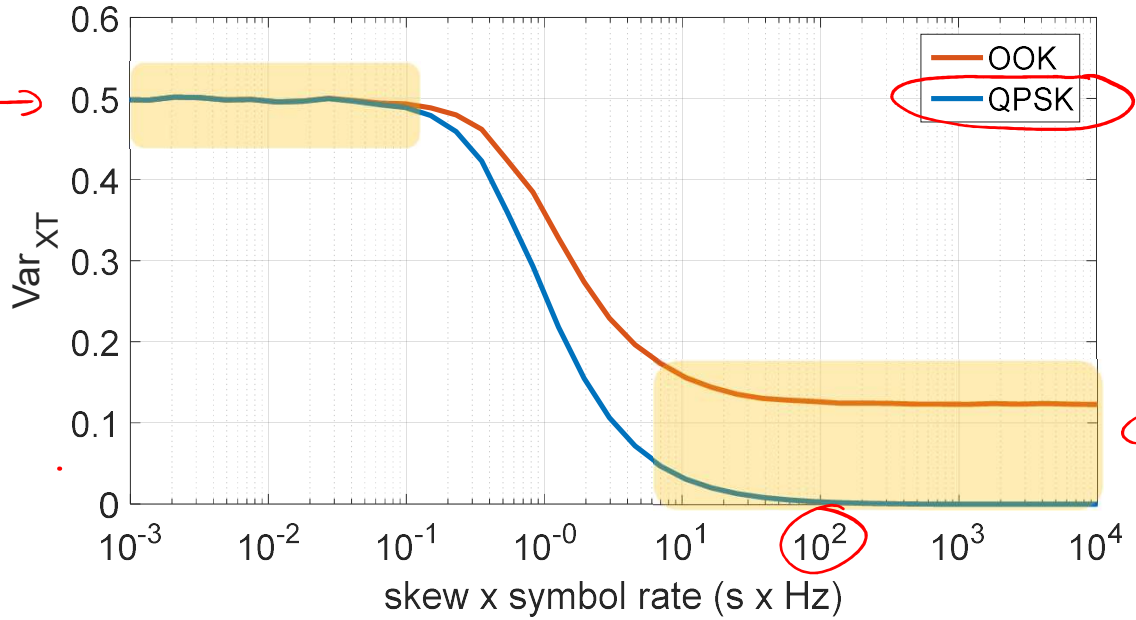
250 ns skew



Variance of average crosstalk power

University of Stuttgart
Germany

large
$cov(\mathcal{A}_\ell, \mathcal{A}_p)$



small
$cov(\mathcal{A}_\ell, \mathcal{A}_p)$

10 GBaud, 10 ns skew

53.7k 7-core MCF : skew 0 ns ... 20 ns

Spectrum of inter-core crosstalk



2dB/
div



75
GHz



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Germany

Concept Test

Concept test

University of Stuttgart
Germany

Which concept would you use to supress inter channel crosstalk?

A Use refractive index trenches

B Polarization Interleaving

C High symbol rate signals



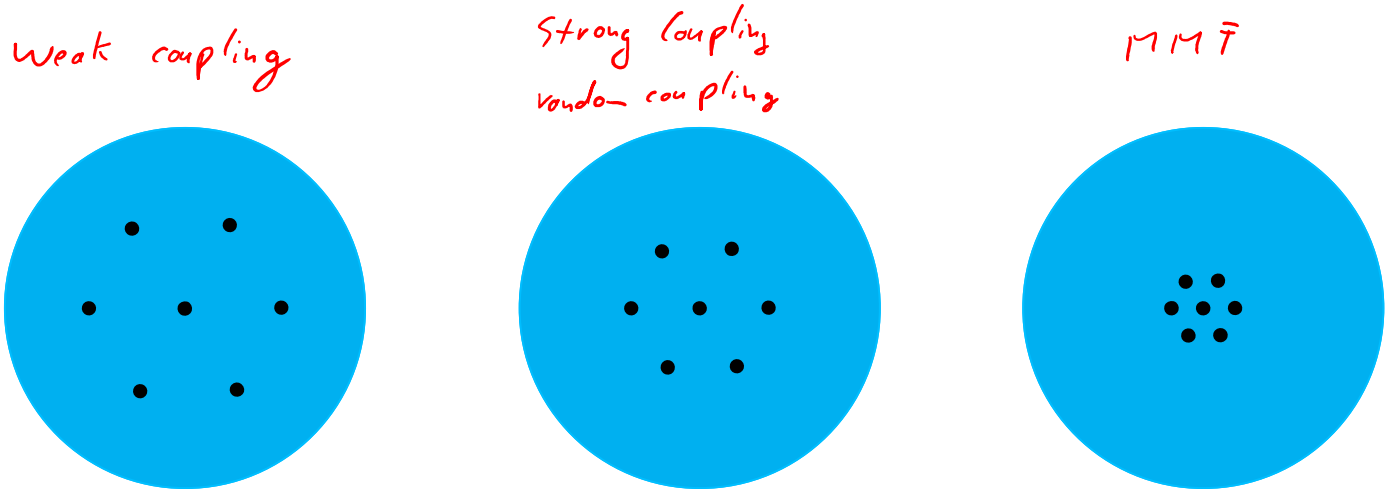
https://ilias3.uni-stuttgart.de/vote/P1CF

Randomly-Coupled Multi-Core Fiber Technology

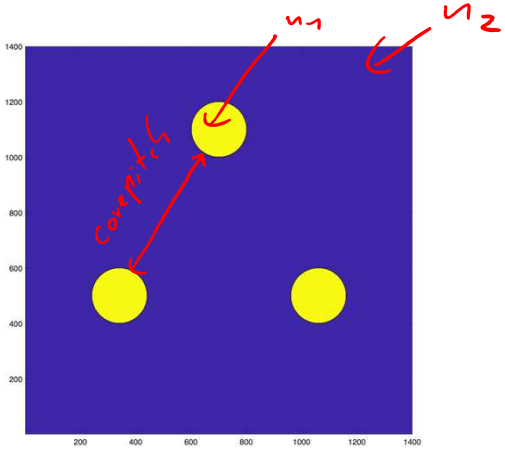
This article considers an even higher density spatially parallel integration technique that allows for random coupling between the cores of multi-core fibers.

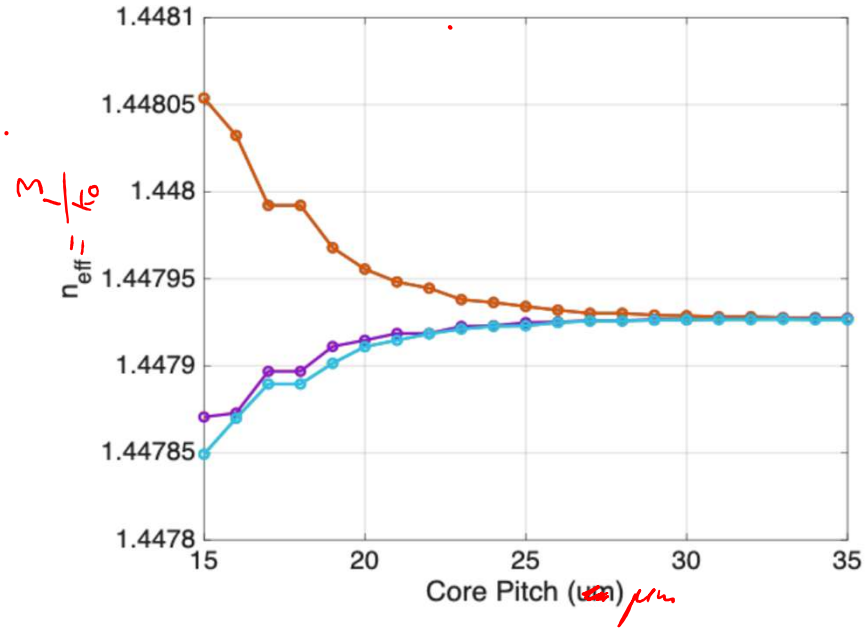
By Tetsuya Hayashi, *Senior Member IEEE*, Taiji Sakamoto, *Member IEEE*, Yusuke Yamada, Roland Ryf, *Fellow IEEE*, René-Jean Essiambre, *Fellow IEEE*, Nicolas Fontaine, *Senior Member IEEE*, Mikael Mazur, *Member IEEE*, Haoshuo Chen, *Senior Member IEEE*, and Takemi Hasegawa, *Member IEEE*

Properties for CC-MCF

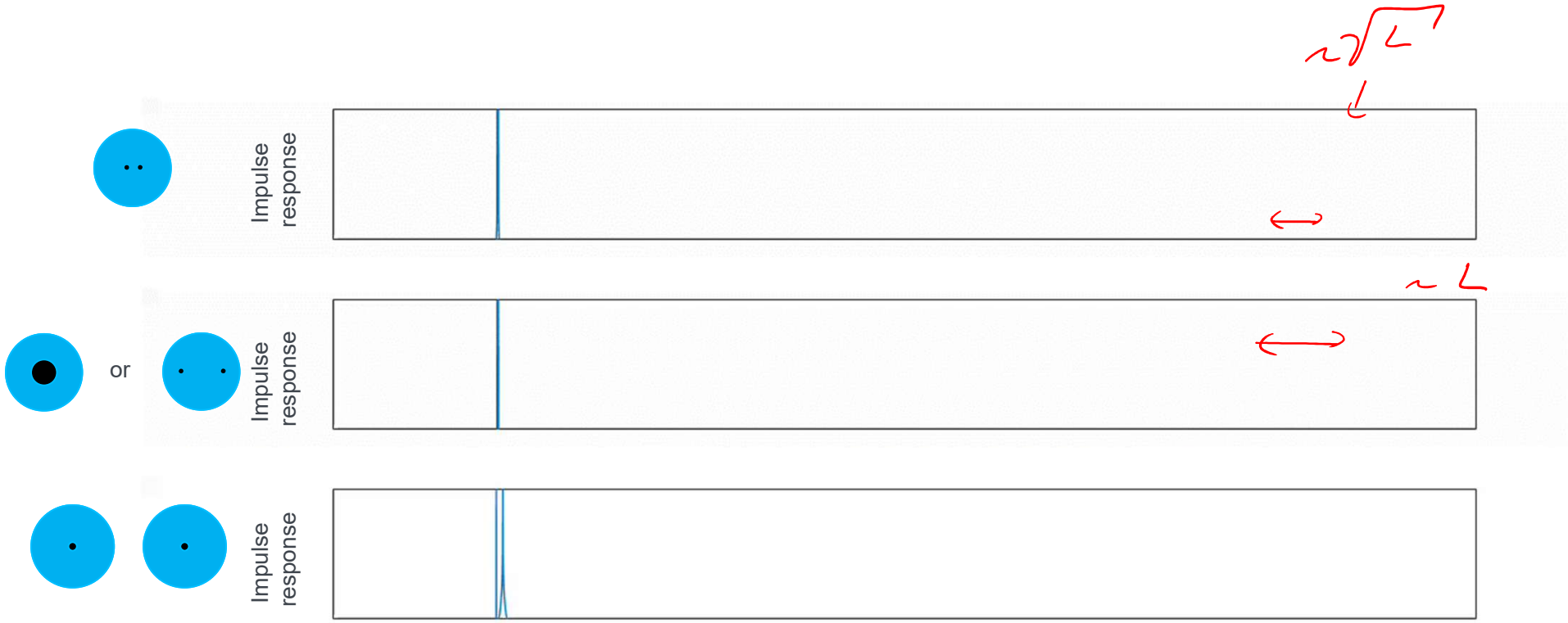


Properties for CC-MCF





Temporal spread of signals in SDM fibers

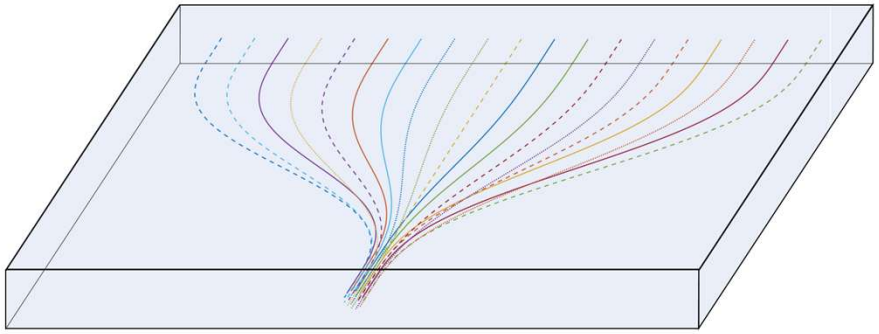


Randomly Coupled 19-Core Multi-Core Fiber with Standard Cladding Diameter

Georg Rademacher[(1)], Menno van den Hout[(1,2)], Ruben S. Luís[(1)], Benjamin J. Puttnam[(1)], Giammarco Di Sciullo[(1,3)], Tetsuya Hayashi[(4)], Ayumi Inoue[(4)], Takuji Nagashima[(4)] Simon Gross[(5)], Andrew Ross-Adams[(6)], Michael J. Withford[(6)], Jun Sakaguchi[(1)], Cristian Antonelli[(3)], Chigo Okonkwo[(2)], Hideaki Furukawa[(1)]

[(1)] NICT, 4-2-1, Nukui-Kitamachi, Koganei, Tokyo, 184-8795, Japan,
[(2)] High Capacity Optical Transmission Lab, Eindhoven University of Technology, Eindhoven, The Netherlands
[(3)] University of L'Aquila and CNIT, 67100, L'Aquila, Italy
[(4)] Sumitomo Electric Industries, Ltd., 1 Taya-cho, Sakae-ku, Yokohama 244-8588, Japan
[(5)] MQ Photonics Research Centre, School of Engineering, Macquarie Univ., Sydney, Australia.
[(6)] MQ Photonics Research Centre, School of Math. and Physical Sciences, Macquarie Univ., Sydney, Australia.
georg.rademacher@nict.go.jp

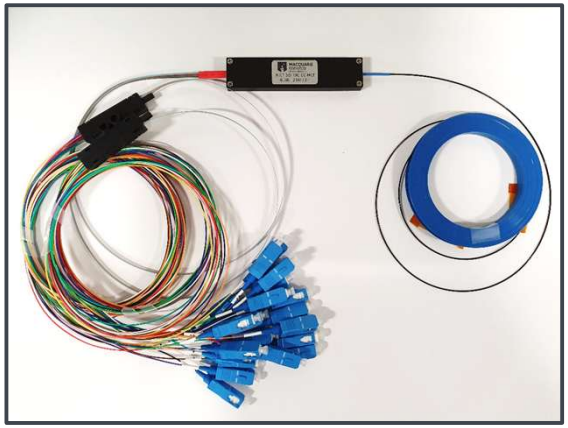
Input single mode fiber array



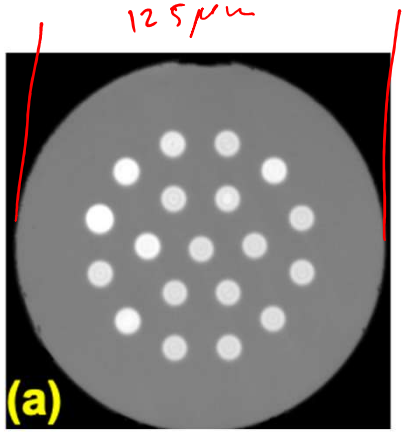
Output RC-19-MCF

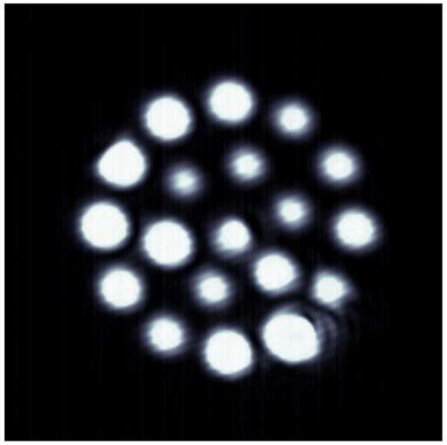
S. Gross and M.J. Withford, "Ultra fast-laser-inscribed 3D integrated photonics," Nanophotonics 4 (1) (2015)

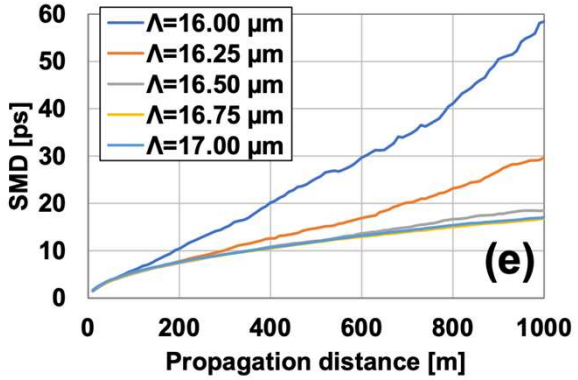


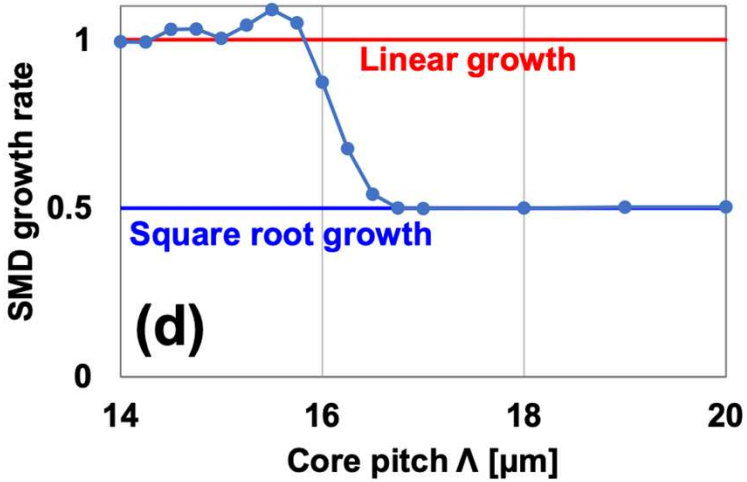


Geo
rg

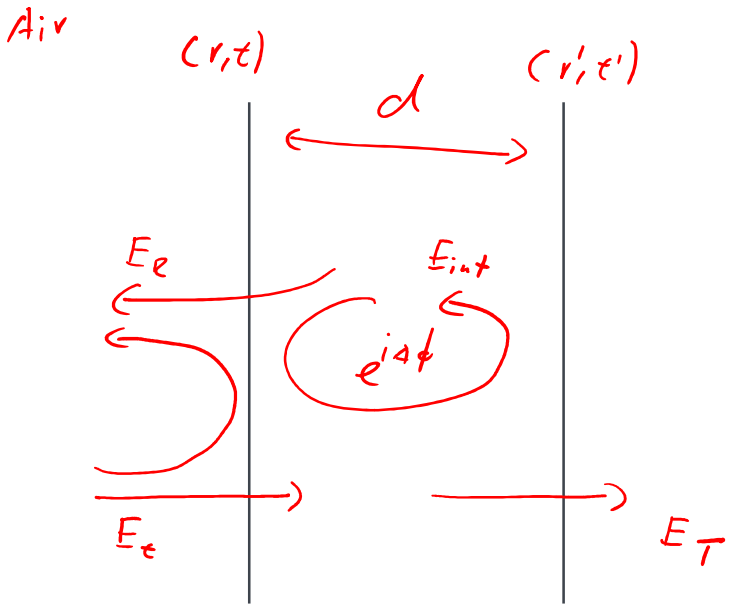

125 µm
(a)




Λ=16.00 µm
Λ=16.25 µm
Λ=16.50 µm
Λ=16.75 µm
Λ=17.00 µm
SMD [ps]
60
50
40
30
20
10
0
0
200
400
600
800
1000
Propagation distance [m]
(e)


SMD growth rate
1
0.5
0
Linear growth
Square root growth
(d)
14
16
18
20
Core pitch Λ [µm]

Fabry-Perrot interferometer



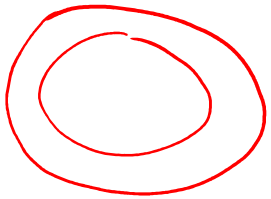
$$\Delta = 2 n d \cos(\theta)$$

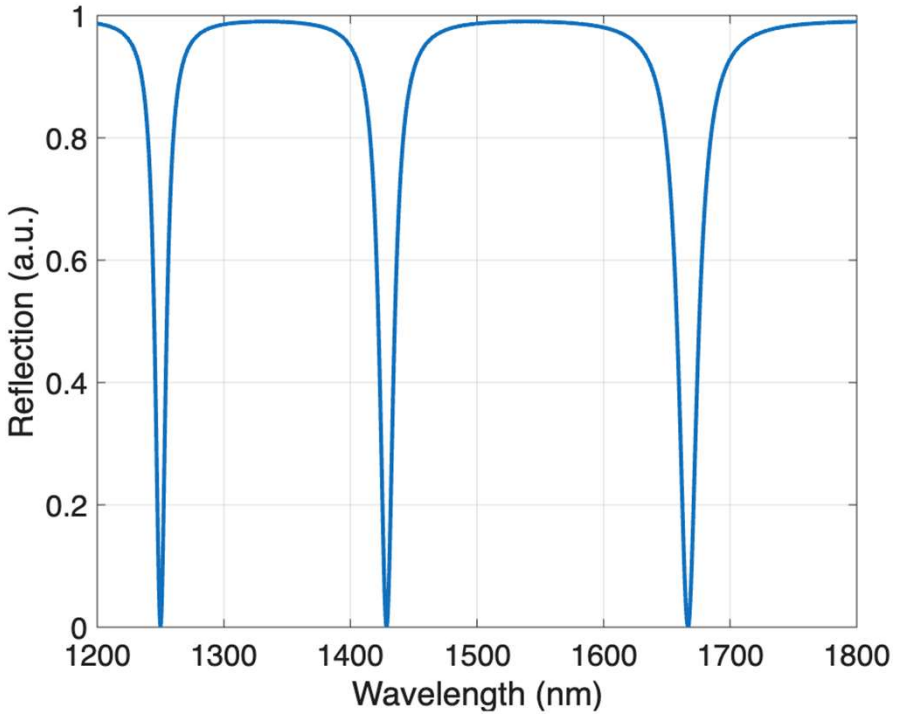
$$\delta = \frac{2\pi}{\lambda} \cdot \Delta = \frac{4\pi n d \cos(\theta)}{\lambda}$$

Fabry-Perrot interferometer

$$I_T = \frac{I_0}{1 + F \sin^2\left(\frac{\Delta\phi}{2}\right)}$$

$$F = \frac{4rr'}{(1-rr')^2}$$





Hollow Core Fiber

nature photonics

Article

https://doi.org/10.1038/s41566-025-01747-5

Broadband optical fibre with an attenuation lower than 0.1 decibel per kilometre

Received: 4 April 2025

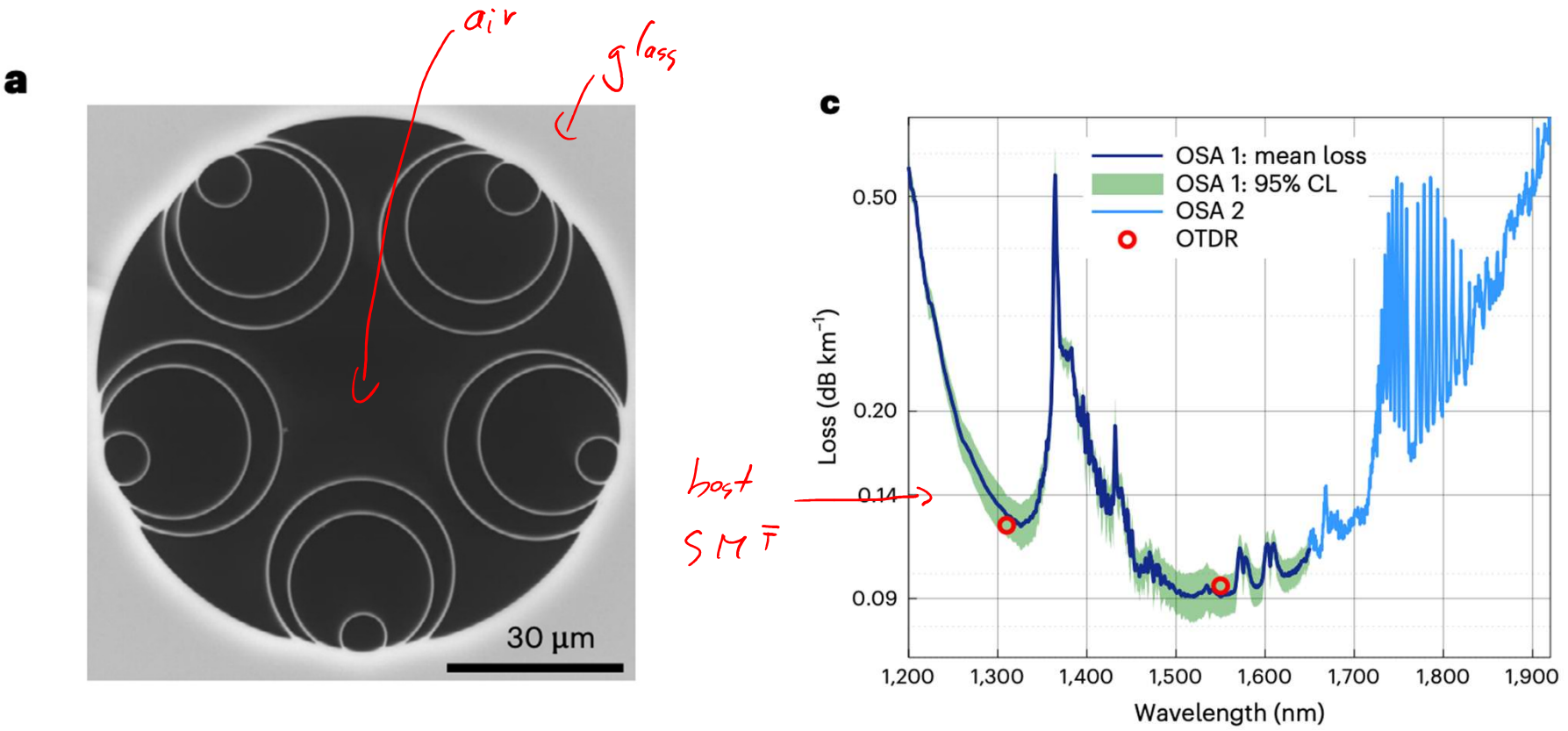
Accepted: 22 July 2025

Published online: 1 September 2025

Check for updates

Marco Petrovich[1,3], Eric Numkam Fokoua[1,3], Yong Chen[1], Hesham Sakr[1], Abubakar Isa Adamu[1], Rosdi Hassan[1], Dong Wu[1], Ron Fatobene Ando[1], Athanasios Papadimopoulos[1], Seyed Reza Sandoghchi[1], Gregory Jasion[2] & Francesco Poletti[1,2]

Hollow Core Fiber



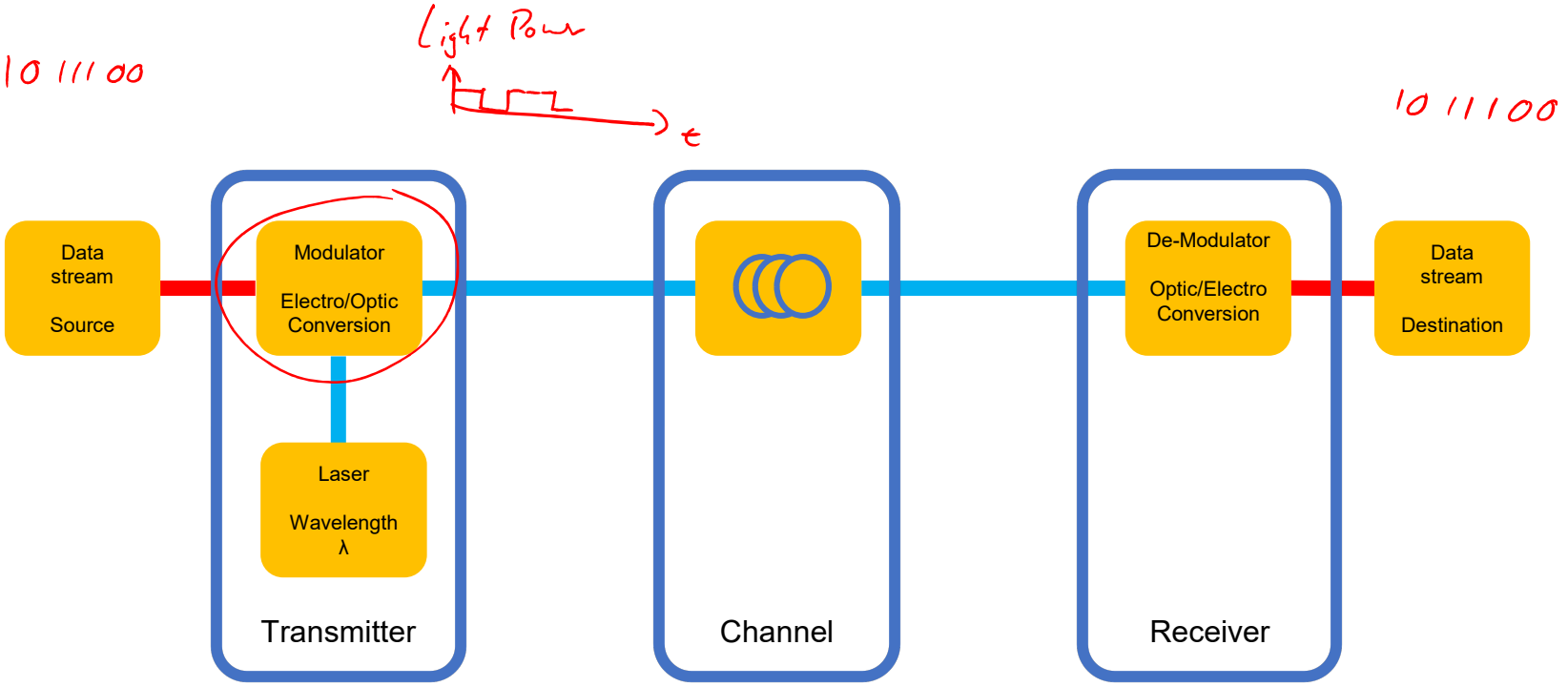


University of Stuttgart
Germany

New Topic!

Principle of an optical fiber transmission link

University of Stuttgart
Germany



Phase modulation: Pockels-effect

University of Stuttgart
Germany

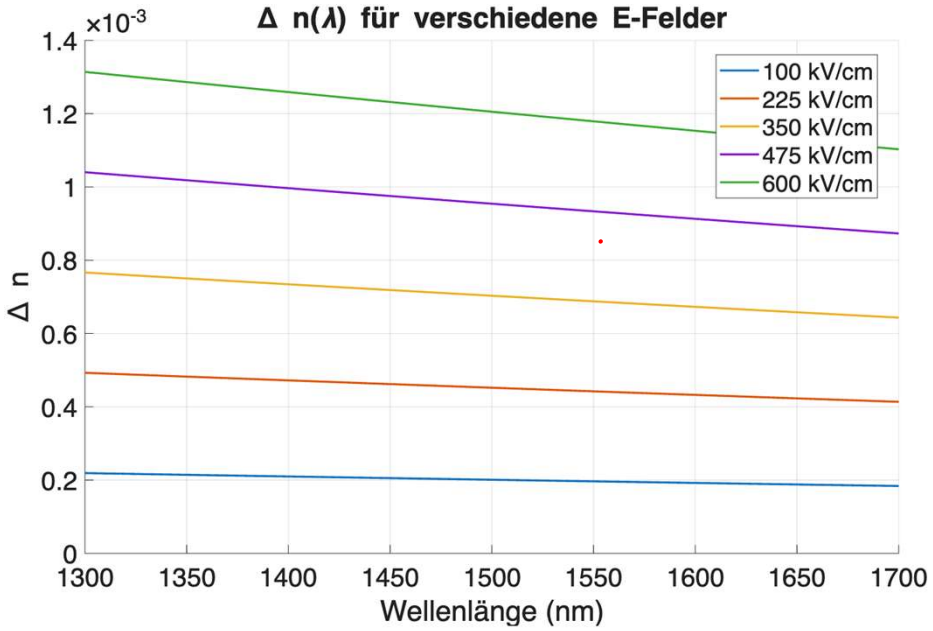
$In\,Ga\,As\,P$

$\Delta\varepsilon \sim E \quad , \quad \varepsilon \sim n^2$

$\Delta n = -n_e^3 \, r_{33} \, \frac{E}{2}$

$n_e = 2.15$

$r_{33} = 31 \cdot 10^{-12}\ m/V$

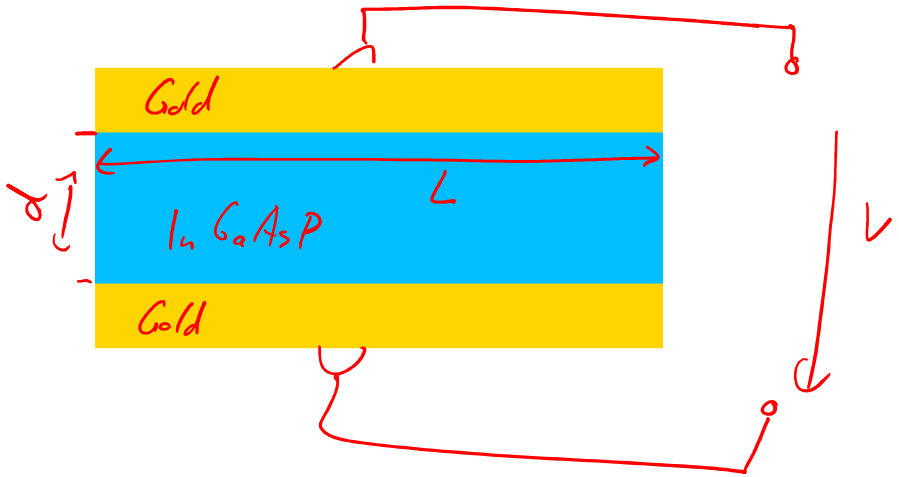


Phase modulation: Pockels-effect

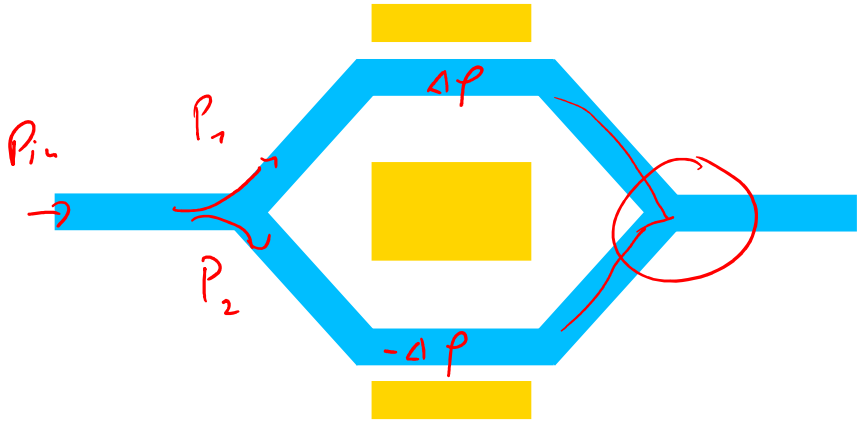
University of Stuttgart
Germany

$$\Delta\varphi = \Delta\beta L = \frac{2\pi}{\lambda}\Delta n_{eff} \cdot L$$

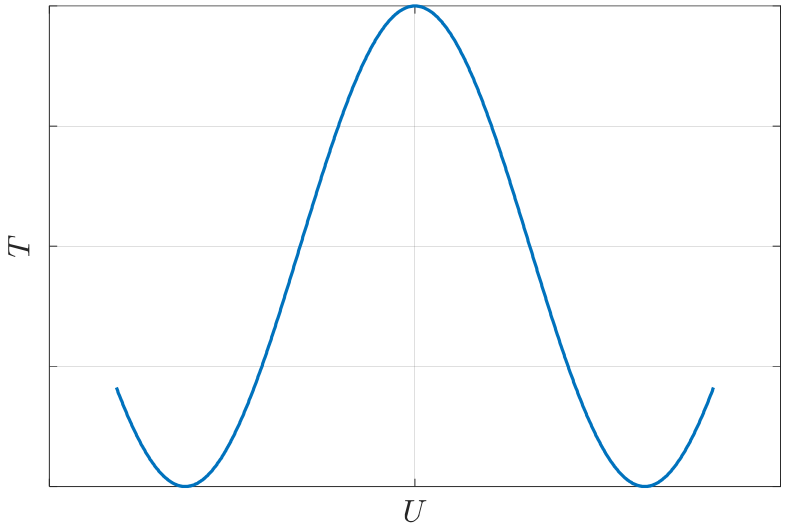
$$\Delta\rho = -\pi n_e^3 r_{33} \cdot \frac{V \cdot L}{d \cdot \lambda}$$



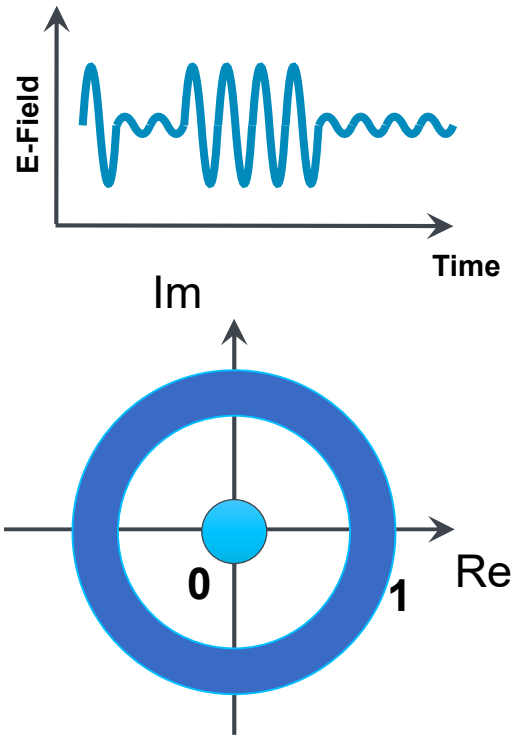
Mach-Zehnder Interferometer

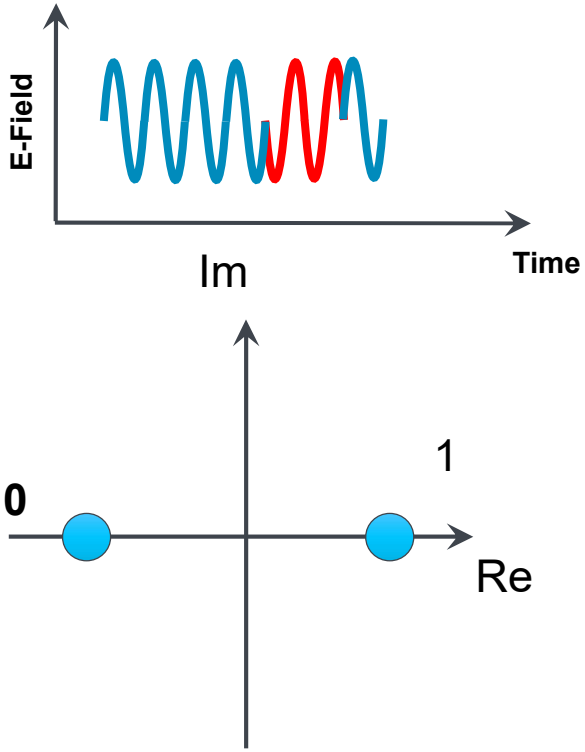


Modulators: Mach Zehnder Interferometer (MZI)

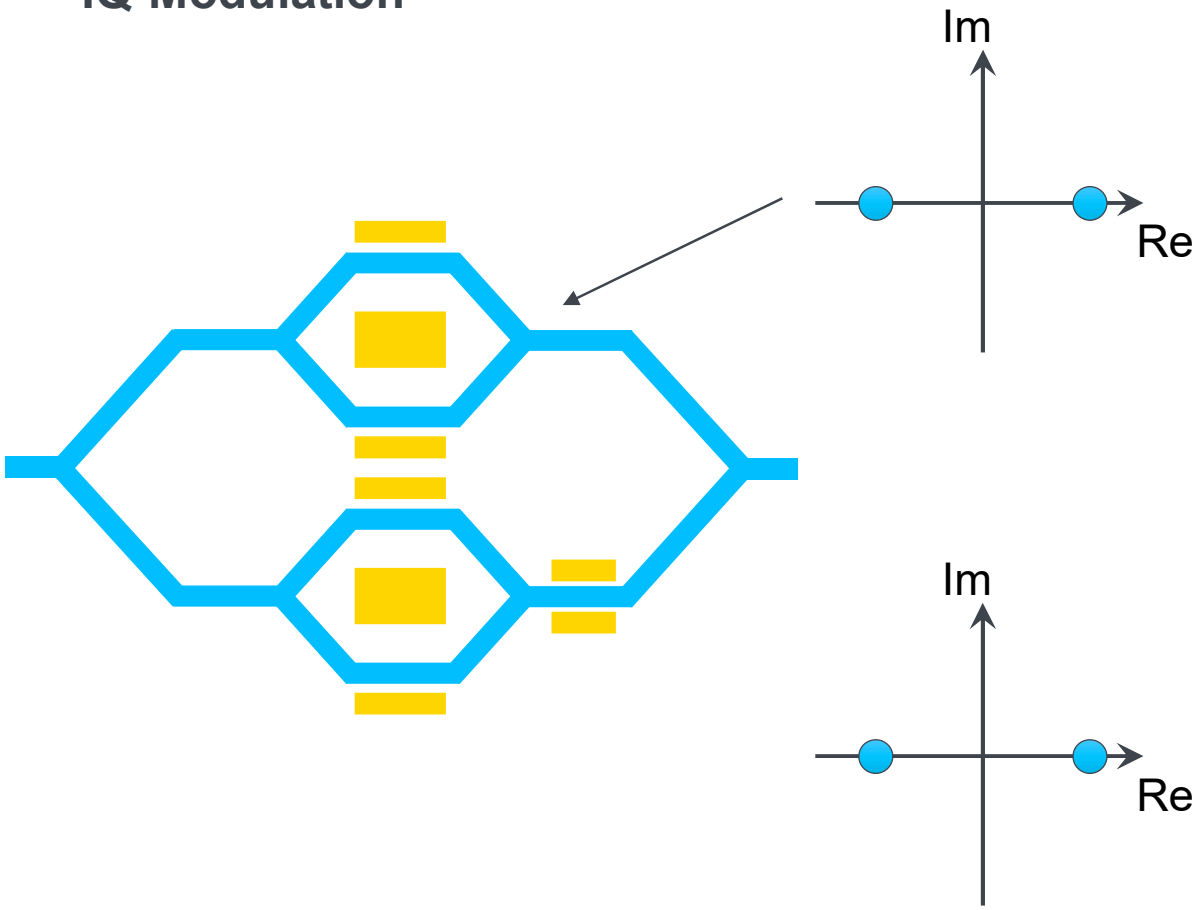


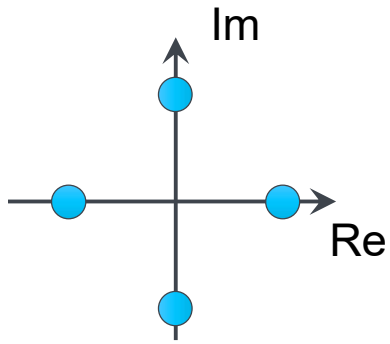
IQ-Modulation



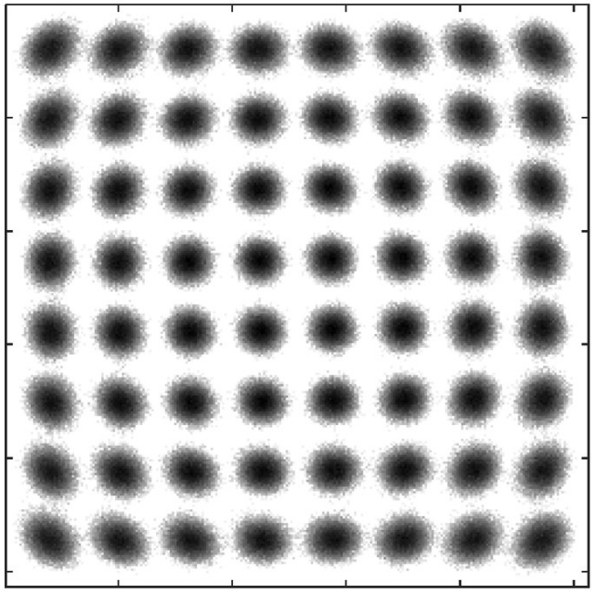


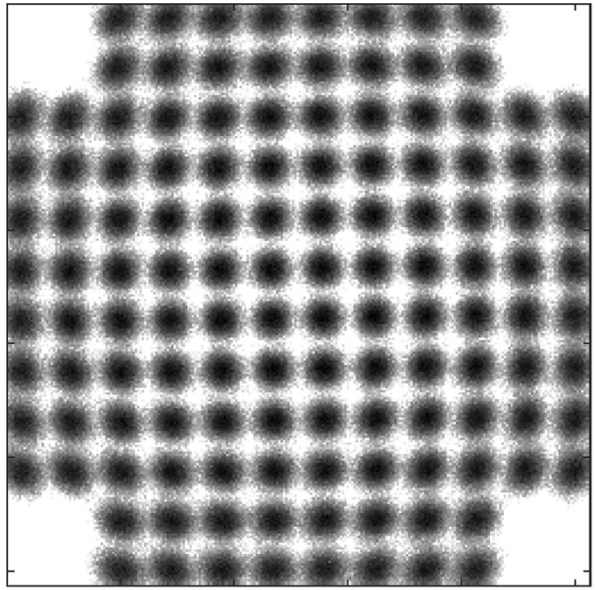
IQ-Modulation

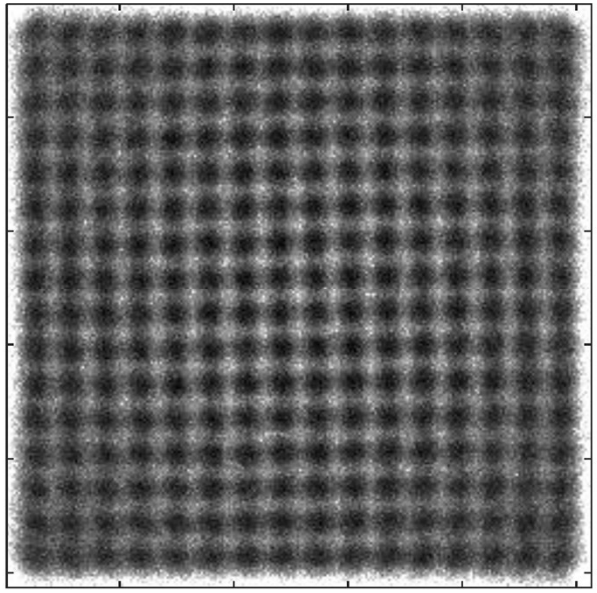




Higher order QAM: Measurements







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Thank you!



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